Laborator 3

3 Fisiere – citire/scriere

Sumar

Files in/out
Sorting
String.split()

3.1 Citire din fisiere text

Java 7 (and newer):

```
new String(Files.readAllBytes(...))

Files.readAllLines(...)

new Scanner(..).hasNextLine(), nextLine()
```

Java 8 (and newer):

```
Files.lines(..).forEach(...)
```

Exemplu care citeste si afiseaza continutul fisierului *infile.txt* cu ajutorul a doua metode:

```
import java.io.IOException;
import java.nio.file.Files;
import java.nio.file.Path;
import java.nio.file.Paths;
import java.util.List;
import java.util.Scanner;

public class Application {
    public static void main(String[] args) {
        try {
            printSmallTextFile("infile.txt");
            System.out.println("\n\n");
            printLargerTextFile("infile.txt");
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    static void printSmallTextFile(String fileName) throws IOException {
        System.out.println("Using Files.readAllLines:");

        Path path = Paths.get(fileName);
        List<String> allLinesInMemory = Files.readAllLines(path);
        for(String line: allLinesInMemory) {
```

```
            System.out.println(": " +line);
        }
    }

    static void printLargerTextFile(String fileName) throws IOException {
        System.out.println("Using Scanner:");
        Path path = Paths.get(fileName);
        try (Scanner scanner =  new Scanner(path)){
            while (scanner.hasNextLine()){
                //process each line in some way
                System.out.println(": " + scanner.nextLine());
            }
        }
    }
}
```

3.2 Scriere in fisier text

Java 7 (and newer)

```
Files.write(path, lines);                                   // overwriting file
Files.write(path, lines, StandardOpenOption.APPEND); // appending to file
BufferedWriter.write(line);
```

Exemple de scriere in fisier:

```
import java.io.BufferedWriter;
import java.io.IOException;
import java.nio.file.Files;
import java.nio.file.Path;
import java.nio.file.Paths;
import java.util.Arrays;
import java.util.List;

public class Application {
    public static void main(String[] args) {
        List<String> lines = Arrays.asList(
            "Lorem ipsum dolor sit amet, vim ne eripuit vulputate." ,
            "Vide tantas dicunt ne ius, mel pertinacia intellegebat ei." ,
            "Sea vidisse aperiam et, eu altera audiam percipitur vis, decore verterem ea nec." ,
            "Veri quando ad est. Ex quidam blandit nec, eos et percipit indoctum.",
            "\n" ,
            "Summo putant consetetur cu vel. Id eum adipisci philosophia, eos commodo principes an." ,
            "Duis eripuit laoreet qui ne, pri maiorum detracto definitiones et." ,
            "Sea quis laudem dolorem eu."
        );
        try {
            System.out.print("Writing outfile1.txt ... ");
            writeSmallTextFile(lines, "outfile1.txt");
            System.out.println("done");

            System.out.print("\n Writing outfile2.txt ... ");
            writeLargerTextFile(lines, "outfile2.txt");
            System.out.println("done");

        } catch (IOException e) {
            e.printStackTrace();
        }
    }
```

```
    static void writeSmallTextFile(List<String> lines, String fileName) throws IOException {
        Path path = Paths.get(fileName);
        Files.write(path, lines);
    }


    static void writeLargerTextFile(List<String> lines, String fileName) throws IOException {
        Path path = Paths.get(fileName);
        try (BufferedWriter writer = Files.newBufferedWriter(path)){
            for(String line : lines){
                writer.write(line);
                writer.newLine();
            }
        }
    }
}
```

3.3 split()

Metoda split() a oricarei instante String va determina extragerea unui array de String-uri rezultat prin extragerea sirurilor de caractere dintre aparitiile separatorului. Separatorul poate fi un sir de caractere, sau o expresie regulata:

```
public class Application {
    public static void main(String[] args) {
        String line = "Lorem    ipsum dolor     sit amet, vim ne eripuit vulputate.\n" +
            "Vide tantas dicunt ne ius, mel pertinacia intellegebat ei." ;

        System.out.println("Split by one space char: ");
        String[] words = line.split(" ");
        for(String w: words) {
            System.out.print("[" + w + "]");
        }

        System.out.println("\nSplit by regex \\s+: ");
        String[] words2 = line.split("\\s+");
        for(String w: words2) {
            System.out.print("[" + w + "]");
        }
    }
}
```

3.4 Sortare String-uri

3.4.1 Folosind Arrays.sort

```
    String[] words3 = {"Lorem","ipsum","dolor"," sit","amet",",","vim","ne","eripui"};
    System.out.println("\nSort 1, array");
    Arrays.sort(words3);
    for(String w: words3) {
        System.out.print("[" + w + "]");
    }
    System.out.println();
```

3.4.2 Folosind Collections.sort

```
System.out.println("\nSort 2, Collection");
List<String> myStringList = Arrays.asList(words3);
Collections.sort(myStringList);
for(String w: myStringList) {
    System.out.print("[" + w + "]");
}
System.out.println();
```

3.4.3 Folosind TreeSet

```
System.out.println("\nSort 3, TreeSet");
Set<String> myTreeSet = new TreeSet<>(myStringList);
for(String w: myTreeSet) {
    System.out.print("[" + w + "]");
}
```

3.5 Probleme

3.5.1 Deschideti proiectul LaboratoareSoftware. Creati un nou subpachet (ex. lab3).
Adaugati textul de mai jos in fisierul **in.txt**, si apoi scrieti cod pentru a cititi datele din fisier intr-o lista in memorie (folositi metoda split)

a) parcurgeti fiecare linie si adaugati inca un new line (caracterul \n) pentru fiecare sfarsit de linie, apoi afisati
b) parcurgeti fiecare linie si adaugati un newline (caracterul \n) pentru fiecare caracter punct (.) si apoi afisati
c) salvati in fisierul **out.txt** rezultatul modificarilor de la a) si b)

Textul este:
Lorem ipsum dolor sit amet, vim ne eripuit vulputate. Vide tantas dicunt ne ius, mel pertinacia intellegebat ei. Sea vidisse aperiam et, eu altera audiam percipitur vis, decore verterem ea nec. Veri quando ad est. Ex quidam blandit nec, eos et percipit indoctum.
Summo putant consetetur cu vel. Id eum adipisci philosophia, eos commodo principes an. Duis eripuit laoreet qui ne, pri maiorum detracto definitiones et. Sea quis laudem dolorem eu.

3.5.2 Deschideti proiectul **Students** de data trecuta si:
Citit studenții din fișierul **studenti_in.txt** cu câmpuri separate cu virgulă. Câte un student pe linie.
Afisati studentii cititi. Salvati lista de studenți în fișierul **studenti_out.txt** sortati după numele de familie (nume).

studenti_in.txt:
1024,Ioan,Mihalcea,ISM141/1
1025,Andrei,Popa,ISM141/2
1026,Bianca,Popescu,TI131/2
1026,Anamaria,Prodan,TI131/1
1027,Paul,Mohanu,TI132/1
1028,Mihai,Ionescu,TI132/2

3.5.3 Tema de casă: Salvati lista citita anterior, sortati după 2 câmpuri: formație de studiu și apoi nume si apoi salvati in fisierul **studenti_out_sorted.txt**

3.6 Referinte aditionale

Reading Files
https://www.w3schools.com/java/java_files_read.asp

Writing Files
https://www.w3schools.com/java/java_files_create.asp

Various methods reading files
https://www.geeksforgeeks.org/different-ways-reading-text-file-java/

File.readAllBytes
https://docs.oracle.com/javase/7/docs/api/java/nio/file/Files.html#readAllBytes%28java.nio.file.Path%29

File.readAllLines
https://docs.oracle.com/javase/7/docs/api/java/nio/file/Files.html#readAllLines%28java.nio.file.Path,%20java.nio.charset.Charset%29

Files.lines
https://docs.oracle.com/javase/8/docs/api/java/nio/file/Files.html#lines-java.nio.file.Path-

Scanner
https://docs.oracle.com/javase/7/docs/api/java/util/Scanner.html

Reading and writing text files
http://www.javapractices.com/topic/TopicAction.do?Id=42